

A pan-dental-source in-vitro osteogenic program and a human single-cell audit of its in-vivo transferability

1 · IN-VITRO OSTEOGENIC PROGRAM (validated deliverable)

RRA meta-analysis across 5 clean dental-MSC sources (PDLSC·DPSC·GSC·SHED·DFC) + leave-one-out stability + WGCNA core

744 LOO-stable genes (350./394.) · 52-gene core · separable in vitro (GSE316449, AUC=1.0 but underpowered)



2 · HUMAN SINGLE-CELL TRANSFERABILITY AUDIT (the load-bearing result)

GSE303003 human jaw: 4 MRONJ vs 4 cyst (65,454 cells) + GSE104473 mouse distraction osteogenesis

. data validated:
osteoclast/myeloid
expansion ($p=0.029$)

. NOT suppressed in MRONJ
mesenchyme (sign test 43.9%,
below chance)

. NO osteoprogenitor home:
program highest in myeloid,
lowest in mesenchymal
(osteo-markers positive ctrl peak there)

. a reproducible in-vitro program need not correspond to any discrete in-vivo osteoprogenitor state (mouse + human converge)



3 · EXPLORATORY GENETIC + REPURPOSING (no robust nomination)

Druggable-genome MR (lead-cis-SNP Wald + full-panel IVW/MR-Egger) vs heel-eBMD; FinnGen osteoporosis replication

GREM2 = directional-pleiotropy artifact (Egger $p=6e-11$) · only weak SRGN survives · no robust druggable target

Connectivity map (4 methods): dexamethasone positive control + HDAC inhibitors — methodological sanity check

Honest scope: reproducible in-vitro resource + human-anchored lesson that in-vitro transcriptomic reproducibility does not imply in-vivo cellular correspondence.
ISG culture signal (incl. core hub EPST11) persists through RRA+LOO+WGCNA. No therapeutic or mechanistic claim.